

ДОМАШНЕЕ ЗАДАНИЕ ПО ЭЛЕМЕНТАРНЫМ ФУНКЦИЯМ.

Для всех факультетов, кроме ФН

1. Найти область определения функции.
2. Исследовать на четность – нечетность данную функцию.
3. Построить эскизы графиков следующих элементарных функций (а – е). Асимптоты графиков укажите пунктиром в той же координатной плоскости.

№	Вариант 1	Вариант 2	Вариант 3	Вариант 4
1	$y = \sqrt{\frac{2x+1}{x-3}}$	$y = \frac{\sqrt{x^2-5x+6}}{x+4}$	$y = \sqrt{\frac{2x}{x+2}} - 1$	$y = \frac{x-1}{\sqrt{x^2+2x-3}}$
2	$y = x^4 - \frac{5}{x^6}$	$y = x^3 + \frac{2}{x^5}$	$y = \frac{x+1}{x^2-9}$	$y = \frac{1}{3^x + 3^{-x}}$
3(а)	$y = \sqrt{3x+1} - 2$	$y = x^2 + 5x + 6 $	$y = (x-2)^3 + 1$	$y = (2x+1)^3$
3(б)	$y = \left \frac{3x+1}{x+1} \right $	$y = \frac{x}{x-1}$	$y = \frac{2x-1}{x-1}$	$y = \frac{4 x -1}{x+2}$
3(в)	$y = 2^{ x -1} - 4$	$y = 3^{1- x } - 1$	$y = \left(\frac{1}{2}\right)^{2x-1}$	$y = 2^{\frac{1}{2}x+1}$
3(г)	$y = \log_2 \frac{2}{x-1}$	$y = \log_3 \frac{1}{x+1}$	$y = \log_{\frac{1}{2}} (x -1) + 2$	$y = \log_2 (3x-1) $
3(д) *	$y = 2 \sin\left(3x - \frac{\pi}{2}\right)$	$y = 2 \operatorname{tg} \frac{x}{3} + 1$	$y = 3 \operatorname{ctg} \frac{x}{2} - 1$	$y = 3 \cos\left(2x - \frac{\pi}{2}\right)$
3(е) **	$y = \frac{\pi}{2} - \operatorname{arctg} x$	$y = \frac{\pi}{2} + \arcsin(-x)$	$y = \arcsin(2x-3) $	$y = \frac{\pi}{4} + 2 \operatorname{arctg} x$
№	Вариант 5	Вариант 6	Вариант 7	Вариант 8
1	$y = \sqrt{\frac{x}{2^{x+1}}} - 4$	$y = \log_3 \frac{3x+1}{x-2}$	$y = \sqrt{\log_2 \frac{x+1}{x+2}} - 1$	$y = \sqrt{\left(\frac{1}{3}\right)^{x^2-5x}} - 81$
2	$y = \frac{1}{2^x - 2^{-x}}$	$y = x^2 \sin x$	$y = \frac{x}{x^2+1}$	$y = x^2 + 5x + 6$
3(а)	$y = x^2 - 6x + 5 $	$y = \sqrt{2x+1} - 1$	$y = 1 - \sqrt[3]{x+2}$	$y = 2\sqrt{x-1} + 1$
3(б)	$y = \frac{2x+3}{ x -2}$	$y = \frac{2x-1}{x+1}$	$y = \frac{2x+1}{x-1}$	$y = \frac{ x -1}{3 x +1}$
3(в)	$y = 3^{1-2x}$	$y = 1 - 3^x $	$y = 2 \cdot 3^{1-\frac{1}{2}x}$	$y = 4 \cdot 2^x - 1 $
3(г)	$y = \log_5 \frac{x-2}{5}$	$y = \log_{\frac{1}{3}} (3x-1)$	$y = \log_{1/3} (x+3) - 1$	$y = \log_2 \frac{x+1}{2}$
3(д) *	$y = 1 + 2 \cos \frac{x}{2}$	$y = 1 + \sin 2x $	$y = 2 \cos\left(\frac{\pi}{2} - x\right) + 1$	$y = 2 \sin(\pi - x) - 2$
3(е) **	$y = \arcsin\left(1 - \frac{x}{3}\right)$	$y = -\arccos(1-x)$	$y = \frac{\pi}{6} + \arcsin(x +1)$	$y = \frac{1}{2} \operatorname{arctg}(x+1)$

№	Вариант 9	Вариант 10	Вариант 11	Вариант 12
1	$y = \sqrt{x(x^2 - 6x + 9)}$	$y = \arcsin \frac{x^2 + 5x}{6}$	$y = \log_{\frac{1}{2}} \frac{x+1}{x-2}$	$y = \arccos \frac{x^2 - 5x}{6}$
2	$y = x + \operatorname{tg} x$	$y = \log_2 \frac{x-1}{x+1}$	$y = \sqrt{1+x} + \sqrt{1-x}$	$y = \sqrt[3]{3+x} + \sqrt[3]{3-x}$
3(а)	$y = -x^2 + 4x - 3 $	$y = \sqrt[3]{3-2x}$	$y = 3 - \sqrt{x+4}$	$y = (x+2)^3 - 1 $
3(б)	$y = \frac{x-1}{x+1}$	$y = \frac{ x }{x+2}$	$y = \left \frac{3x+1}{2-x} \right $	$y = \frac{x-1}{x-2}$
3(в)	$y = 2 \cdot \left(\frac{1}{5} \right)^{2x+1}$	$y = 2^{3-2x}$	$y = 3 \cdot 2^{-x} - 1$	$y = 3 \cdot 2^{- x }$
3(г)	$y = \log_{\frac{1}{2}} (x - 1) + 2$	$y = \log_2 \frac{2x+1}{4}$	$y = \lg(2 x - 5)$	$y = \log_{\frac{2}{3}} (2x - 3) + 1$
3(д) *	$y = \frac{3}{2} \sin \left(x - \frac{\pi}{4} \right) - \frac{3}{2}$	$y = \frac{1}{2} \left \cos \left(x + \frac{\pi}{4} \right) \right $	$y = 1 - \operatorname{tg} \left(x + \frac{\pi}{4} \right)$	$y = 2 \operatorname{tg} \left(x - \frac{\pi}{3} \right)$
3(е) **	$y = \frac{\pi}{4} + \arccos 2x$	$y = -\frac{\pi}{3} + \arccos \frac{x}{3}$	$y = \frac{\pi}{4} + 3 \arcsin x$	$y = \frac{1}{3} \operatorname{arctg}(1-x)$
№	Вариант 13	Вариант 14	Вариант 15	Вариант 16
1	$y = \sqrt{\frac{x+1}{5-2x}}$	$y = \sqrt{\frac{2x-1}{x+3}}$	$y = \frac{\log_3(x+2)}{1 + \sqrt{x^2 - 2x - 3}}$	$y = \sqrt{x - \frac{4}{x}}$
2	$y = \frac{x}{x^2 - 1}$	$y = 5^{x^2} - \frac{1}{x}$	$y = x \cdot \operatorname{tg} x$	$y = x \cdot \arcsin x$
3(а)	$y = \sqrt{(x+1)^3} - 2$	$y = \sqrt{3 - \frac{1}{2}x}$	$y = (2x-1)^3 + 1$	$y = 2 - \sqrt{x-1}$
3(б)	$y = \frac{x+2}{3x-4}$	$y = \frac{2x}{ x -1}$	$y = \left \frac{3x-5}{x-1} \right $	$y = \left \frac{1-x}{2x+1} \right $
3(в)	$y = 2^{x-1} - 4 $	$y = 2 - 2^{-x}$	$y = \frac{1}{2} \cdot 2^{ x } - 3$	$y = 2 \cdot \left(\frac{2}{3} \right)^{ x +1}$
3(г)	$y = \log_3 \left(\frac{9}{ x +1} \right)$	$y = \log_2(2x+4) $	$y = \log_3(4x+1) - 2$	$y = \log_2(2x-1)^3$
3(д) *	$y = \frac{1}{4} + \frac{1}{2} \cdot \sin \frac{x}{2}$	$y = 1 - \cos 2x$	$y = \frac{1}{2} \operatorname{tg} 3x - \frac{\sqrt{3}}{2}$	$y = \frac{1}{2} \operatorname{ctg} 2x + 1$
3(е) **	$y = 3 \operatorname{arctg}(1+2x)$	$y = 2 \operatorname{arctg}(x+1)$	$y = -\arccos(2x-1)$	$y = \frac{1}{2} \arccos(3x+1)$

№	Вариант 17	Вариант 18	Вариант 19	Вариант 20
1	$y = \frac{1 + \sqrt{x-1}}{\sqrt{4-x^2}}$	$y = \sqrt{\log_3(2-x)}$	$y = \sqrt{\log_2 \cos x}$	$y = \frac{\log_3(1-x^2)}{1 + \sqrt{2x-1}}$
2	$y = x^2 \operatorname{arctg} x$	$y = \frac{2^x + 2^{-x}}{3^x + 3^{-x}}$	$y = 3^{\sqrt{x^2-4}} + 1$	$y = \frac{3^x + 3^{-x}}{4^x - 4^{-x}}$
3(а)	$y = (x - 1)^3 + 1$	$y = 1 - (x - 1)^3$	$y = \sqrt[3]{2x-1} - 1$	$y = \sqrt[3]{ x - 2} + 1$
3(б)	$y = \frac{x-1}{3x+1}$	$y = \frac{3x+1}{x-1}$	$y = \left \frac{x+1}{2x-1} \right $	$y = \frac{x+4}{2x+6}$
3(в)	$y = 5^{x+1} - 1 $	$y = 2 \cdot 3^{-x} - 3$	$y = 2^{3-2x} + 4$	$y = 4^{-x+1} - 2$
3(г)	$y = 1 - \log_3(x-2)$	$y = \log_3(3 x - 6)$	$y = 1 - \log_{2/5}(x+1)$	$y = \log_{\frac{1}{3}}(x-3) + 1$
3(д) *	$y = \frac{1}{3} \cdot \operatorname{tg} 2x$	$y = 1 - 2 \cos x $	$y = 1 + 2 \cos \frac{x}{3}$	$y = 1 + 3 \cdot \sin \frac{x}{3}$
3(е) **	$y = -\frac{\pi}{2} + \arccos 2x$	$y = \arcsin(x-1) + \frac{\pi}{2}$	$y = \frac{\pi}{3} + \arcsin(x -1)$	$y = \pi - \arccos(x+1)$
№	Вариант 21	Вариант 22	Вариант 23	Вариант 24
1	$y = \log_2(3-2x-x^2)$	$y = \sqrt{\sin 2x-1}$	$y = \sqrt{3 \cdot 2^x - 4^x} - 2$	$y = \sqrt{\sin 3x-1}$
2	$y = 2^{\sqrt{1-x^2}}$	$y = \frac{\sqrt{x^2+1}}{x}$	$y = x \cdot \sqrt[3]{x-1}$	$y = \frac{3x^2+2}{x-1}$
3(а)	$y = 1 - \sqrt{x-2}$	$y = (2x-1)^3 + 2$	$y = -1 + \sqrt{4x+3}$	$y = 3 - \sqrt{x+4}$
3(б)	$y = \frac{2x-1}{2-x}$	$y = \frac{3x}{2-x}$	$y = \frac{x-2}{x+1}$	$y = \frac{2x}{ x -1}$
3(в)	$y = 1 - 3^{x+1} $	$y = 2^{ x +2} - 4$	$y = 1 - 3^{x+1}$	$y = 2^{x+3} - 1 $
3(г)	$y = \log_3(3x+6)$	$y = \left \log_{1/2}(x+1) \right ^3$	$y = 1 + \log_5(2x-1)$	$y = \log_2(4-2x)$
3(д) *	$y = 2 \sin 3x - 1$	$y = 3 \operatorname{tg} \frac{x}{2} - \sqrt{3}$	$y = 4 \sin \left(\frac{x}{2} + \frac{\pi}{2} \right)$	$y = 3 \cdot \cos \left(\frac{x}{3} - \frac{\pi}{3} \right)$
3(е) **	$y = 2 \operatorname{arctg}(x -1)$	$y = \frac{\pi}{3} - 2 \arccos x$	$y = \left \frac{\pi}{3} + \operatorname{arctg} 2x \right $	$y = \frac{1}{2} \operatorname{arctg} 2x - \frac{\pi}{4}$

№	Вариант 25	Вариант 26	Вариант 27	Вариант 28
1	$y = \sqrt{4 - \left(\frac{1}{2}\right)^{x+1}}$	$y = \sqrt{x - \frac{3}{x}}$	$y = \sqrt{x^4 - 5x^2 + 4}$	$y = \sqrt{x} \cdot \lg(9 - x^2)$
2	$y = \frac{x^2}{\sin x}$	$y = \sqrt{4 - x^2} + 3x$	$y = 4x - \frac{3}{x}$	$y = \sqrt{x} \cdot \cos x$
3(a)	$y = \left (x+1)^3 - 1 \right $	$y = 1 - \sqrt[3]{x+2}$	$y = \left 3x - x^2 + 4 \right $	$y = \sqrt{2x+3} - 1$
3(б)	$y = \frac{3x}{2-x}$	$y = \frac{x+3}{x+1}$	$y = \frac{x-1}{x-3}$	$y = \left \frac{x-1}{x+3} \right $
3(в)	$y = 3 - 3^{x+1}$	$y = 1 + 3^{-x+1}$	$y = 3 \cdot 2^{1+\frac{x}{2}}$	$y = \left(\frac{1}{3}\right)^{\frac{x}{2}-1} - 3$
3(г)	$y = \log_3 \frac{ x +1}{9}$	$y = \log_5(5x+1)$	$y = 2 \cdot \log_{\frac{1}{2}}(3x+4)$	$y = 1 - \log_{\frac{1}{3}}(1+x)$
3(д) *	$y = 2 \cos\left(x - \frac{\pi}{4}\right) + \sqrt{3}$	$y = 1 + \sqrt{3} \operatorname{tg}\left(x - \frac{\pi}{3}\right)$	$y = \frac{1}{2} \operatorname{ctg}\left(x - \frac{\pi}{4}\right)$	$y = 3 \cdot \cos\left(x + \frac{\pi}{4}\right)$
3(е) **	$y = \frac{\pi}{2} + 2 \arcsin x$	$y = \arccos(x+1) - \frac{\pi}{3}$	$y = 2 \operatorname{arctg}(x-1) - \frac{\pi}{2}$	$y = 2 \operatorname{arctg} 2x - \frac{\pi}{2}$
№	Вариант 29	Вариант 30	Вариант 31	Вариант 32
1	$y = \frac{1}{\log_3 x} + \sqrt{1-x^2}$	$y = \frac{\sqrt{3-x}}{\ln(2x+1)}$	$y = \frac{\sqrt{3-x}}{\ln(x+2)}$	$y = \ln(3 - 3^{x+1})$
2	$y = x \cdot \sin 3x$	$y = \frac{\operatorname{tg} x}{x^2 - 4}$	$y = \ln\left(\frac{2+3x}{2-3x}\right)$	$y = \sin 3x \cdot \operatorname{ctg} 2x$
3(a)	$y = \sqrt[3]{2x+1} - 1$	$y = 1 + \sqrt[3]{5-x}$	$y = (2x-3)^3 - 5$	$y = 1 - \sqrt{3-x}$
3(б)	$y = \frac{2 x -2}{x+2}$	$y = \left \frac{3x-3}{x+2} \right $	$y = \left \frac{3x-6}{x-1} \right $	$y = \left \frac{4x-4}{3-x} \right $
3(в)	$y = \frac{2}{3^x} - 3$	$y = \left(\frac{1}{2}\right)^{ x -1} - 1$	$y = 2 - \left(\frac{1}{2}\right)^{x+1}$	$y = 3^{1- x }$
3(г)	$y = 1 + \log_2(2x-4)$	$y = \log_{\frac{3}{2}}(3-2x)$	$y = 1 + \log_3(x -2)$	$y = 1 + \log_2(3+2x)$
3(д) *	$y = \sqrt{3} - \operatorname{tg}\left(x + \frac{\pi}{3}\right)$	$y = 1 + \frac{1}{2} \cdot \sin\left(x - \frac{\pi}{2}\right)$	$y = 1 - 2 \cos x$	$y = 1 + \operatorname{ctg}\left(x - \frac{\pi}{4}\right)$
3(е) **	$y = \left \frac{\pi}{4} - \arcsin x \right $	$y = \pi + 2 \operatorname{arctg}(-x)$	$y = 2 \operatorname{arctg}(x+1) - \frac{\pi}{2}$	$y = 2 \arccos(1-x)$